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Predicting depression among public university students in Bangladesh using machine learning models

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Abstract

Background Mental health challenges, particularly depression, anxiety, and stress, are prevalent among university students worldwide. In Bangladesh, public university students face unique stressors, including financial pressure, academic demands, social isolation, and limited access to mental health services. This study aims to identify key risk factors for depression and develop a predictive model to facilitate early intervention.

Methods A cross-sectional study was conducted across 15 public universities in Bangladesh, involving 500 students selected through random sampling. Data were collected via an online survey on socio-demographic, academic, and psychological factors. Depression was assessed using a structured questionnaire developed based on PHQ-9 items and aligned with DSM-5 criteria. Descriptive statistics and chi-square tests were used to explore the distribution and associated factors. Machine learning models (Random Forest, SVM, KNN, and Logistic Regression) were applied. Independent t-tests were performed on precision, recall, accuracy, F1-score, and specificity to determine the best model.

Results The prevalence of depression (85.5%) was higher among students, however, there was no statistical significance ($p = 0.095$) among male students (87.8%) compared to females (81.9%), though. Lower CGPA ($p < 0.001$), financial stress ($p < 0.001$), lack of family support ($p < 0.001$), exposure to ragging ($p < 0.001$), and frequent session jams ($p < 0.001$) were significantly associated with depression. Because of precision, recall, accuracy, F1 score and Specificity; the SVM is best model for the dataset to predict depression. Based on SHAP analysis consultancy service, educational system and session jam were found to be the most influential parameters and addressing them can be a game changer.

Conclusion Depression among public university students in Bangladesh are influenced by academic, financial, and institutional factors. Machine learning models, especially SVM offer promising tools for early depression risk prediction, enabling targeted interventions to improve student well-being.

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Keywords Mental health, University students, Depression, Predictive modeling, Machine learning, Bangladesh

1 Introduction

Mental health issues, including depression, anxiety, and stress, have become significant concerns among university students globally. In the United States, a considerable proportion of undergraduate students suffer from mental health challenges due to academic pressure, social isolation, and financial concerns [1]. These challenges are further compounded by barriers to mental health services, such as stigma and inadequate institutional support [2]. Similarly, research conducted in diverse contexts, including Bangladesh, has revealed alarming rates of depression among university students [3, 4]. Despite these growing concerns, limited research has been conducted specifically on the mental health challenges faced by students in public universities in Bangladesh. Given the unique socio-cultural and economic stressors in this context, there is a critical need for targeted interventions and predictive models that can help identify and support at-risk students.

Public university students in Bangladesh experience distinct mental health challenges compared to their counterparts in private universities [3, 5, 6]. These challenges stem from multiple stressors, including rigorous academic demands, financial burdens, environmental factors, and social pressures, all of which can significantly impact their psychological well-being [5]. While mental health education and early intervention strategies have been explored in various countries, such as peer support networks and digital interventions [7, 8], their applicability in Bangladesh remains largely unexplored. Similar applications of machine learning for depression screening have been reported [9–12], validating our methodological approach.Ethical concerns, such as data privacy and algorithmic bias, further complicate the implementation of such technologies [13]. Addressing this research gap is essential to developing culturally and contextually relevant solutions that can effectively mitigate mental health risks.

This study aims to identify the primary factors contributing to depression among public university students in Bangladesh and to propose actionable, data-driven solutions tailored to their needs. Specifically, it seeks to develop a machine learning-based predictive model to identify students at risk of depression, enabling timely interventions. The initial phase of the research involved an extensive survey among public university students to collect data on academic, psychological, and socioeconomic characteristics, including age, gender, cumulative grade point average (CGPA), financial stress, family status, and social participation. These survey results were analyzed to determine the most significant contributors to depression. Subsequently, machine learning classification models were trained using this data to predict the likelihood of depression among students, with an emphasis on identifying the relative impact of various factors. To validate these findings and explore predictive strategies, a second survey was conducted based on insights from the initial survey and the machine learning outputs.

This research integrates predictive models with actionable recommendations to strengthen mental health support systems in public universities across Bangladesh. It aims to identify key factors influencing student mental health, develop a machine learning-driven prediction interface for at-risk students, and propose targeted interventions

to address these challenges. By establishing a structured framework for mental health support, the study hypothesizes that predictive modeling can significantly improve early detection and intervention, fostering a healthier academic environment. The findings are expected to inform evidence-based policy recommendations for enhancing mental health services in higher education. By leveraging advanced predictive analytics and student-centered strategies, the study contributes to existing knowledge and future interventions in both local and global contexts. Ultimately, this research seeks to create a supportive academic environment that prioritizes mental well-being, enabling students to achieve academic success while maintaining optimal mental health.

2 Methodology

2.1 Study design, setting, and population

This cross-sectional study was conducted between June and October 2024 across 15 public universities in Bangladesh, employing quantitative approaches to assess mental health challenges, particularly depression among undergraduate students. The setting included both urban and semi-urban campuses, representing geographically diverse regions including multiple administrative divisions of Bangladesh. The study framework encompassed data collection, processing, statistical analysis, and predictive model development to identify potential interventions.

2.2 Sampling and participant selection

The minimum sample size (n) was calculated using the formula: $n = Z^2 \times p \times (1-p) / d^2$; where $Z = 1.96$ (95% confidence), $p = 0.50$ (assumed prevalence), and $d = 0.05$ (margin of error), yielding $n \approx 384$. A 50% assumed prevalence was chosen as it provides the maximum sample size under conservative estimation when the true prevalence is unknown. Although recent studies among Bangladeshi university students report higher depression prevalence (≈80–85%) [3], we used 50% as a conservative estimate to maximize sample size when prior estimates vary across populations. This approach is widely recommended for prevalence-based sample size calculations when uncertainty exists. Regarding ML model complexity, while formal power analysis for predictive modeling is still an evolving field, our achieved sample size ($n = 420$) satisfies common rules of thumb such as maintaining > 10 events per variable in logistic regression, and was further supported by the stability of fivefold cross-validation results. Thus, the study was sufficiently powered for exploratory ML applications. To allow for non-response (10%) and ensure adequate statistical power, a total of 420 students ultimately participated in the survey.

According to the University Grants Commission (UGC), Bangladesh has 53 public universities. From these, 15 universities (≈30%) were randomly selected using a simple random sampling approach to ensure institutional diversity. Within these universities, recruitment was conducted by disseminating the survey link (Google Form) through official and widely used student social media groups (e.g., Facebook, WhatsApp). Participation was voluntary and self-administered; therefore, while the selection of universities was random, the participation of students followed an open, non-probabilistic recruitment process. This distinction clarifies that randomization applied only at the institutional level, whereas student-level participation was based on voluntary response. To minimize non-response bias, the participation request was reposted twice at one-week intervals. Inclusion criteria required students to be currently enrolled in a public

university and willing to provide informed consent. Students with incomplete responses or those unwilling to participate were excluded from the study.

2.3 Respondents and data collection tools

Data was collected through an online survey distributed via Google Forms. The survey collected comprehensive information on students' socioeconomic, psychological, and academic characteristics, including variables such as gender, CGPA, financial stress, family support, and relationship status, exposure to ragging, session jams, Depression status and access to mental health services.Depression status was measured using a 5-item scale, which was adapted from the PHQ-9 and DSM-5. The items and item-level response distributions for all depression-related items are given in Supplementary Table 1 and Supplementary Table 2.

Cronbach's alpha of the 5 items is 0.814 (Supplementary Table 3), and Cronbach's alpha is not increased (significantly) with the deletion of any item, that is why, the items are suitable for measuring depression. All items were dichotomous (yes/no). The corrected item-total correlations are 0.435 to 0.614, that is, the items are positively correlated with each other. A score of 2 or more out of 5 indicates moderate or higher depression, consistent with the PHQ-9, where a score of more than 9 out of 27 indicates moderate or higher depression [14].

2.4 Data collection and quality control

To maintain data integrity, all responses were reviewed for completeness before inclusion in the analysis. Missing data were handled by removing affected records. Categorical variables were converted into numerical values for statistical analysis. Strict quality control measures, including duplicate response checks and logical consistency validation, were implemented. Incomplete or logically inconsistent responses were removed before model training to ensure data quality.

2.5 Data analysis

Descriptive statistics were used to summarize the prevalence of depression among students. Inferential statistical techniques, including logistic regression and chi-square tests, were applied to assess the association between different variables and mental health outcomes.While some cells had expected counts < 5, Fisher's Exact Test was not performed due to analytical constraints; findings from such subgroups were interpreted with caution. Machine learning models, including Support Vector Machine (SVM), Random Forest, Logistic Regression, and k-Nearest Neighbors (KNN), were trained using 80% of the data, while the remaining 20% was reserved for model testing. Missing values were removed in this version; however, the user-friendly interface includes logic for handling missing inputs via default population mean/mode imputation. Model performance was evaluated using accuracy, precision, recall, specificity, and F1-score. Formal power analysis was not conducted for ML models; instead, cross-validation and regularization techniques were used to mitigate overfitting.

2.6 Machine learning models

All analyses were conducted in Python 3.8 (Anaconda distribution) using scikit-learn v1.0.2 and SHAP v0.41.0. All numeric features were standardized using z-score

normalization before model training. Hyperparameter was performed via grid search using fivefold cross-validation for each model, optimizing over the following ranges:

Random Forest: n_estimators {100,200,500}, max_depth {None,10,20}, min_samples_split {2,5}.

SVM: C {0.1,1,10}, kernel {'linear','rbf'}, gamma {'scale','auto'}

KNN: n_neighbors {3,5,7}, weights {'uniform','distance'}

Logistic Regression: C {0.1,1,10}, penalty {'l2'}

The final optimal hyperparameters selected for each model after grid search and used to generate the reported results are summarized in Table 4. All performance metrics presented in Tables 2 correspond to these optimal configurations.

2.7 Model training and evaluation

For the binary classification task, the following machine learning models were selected: Support Vector Machine (SVM), Random Forest, Logistic Regression, and k-Nearest Neighbors (KNN) (Fig. 1). For training the models, 80% of the labeled data were used while the remaining 20% were reserved for testing the trained models.

We set the random seed to 42 for all randomized operations, including data splitting and model initialization, which ensures the reproducibility of results and consistency across different runs and model comparisons. The same fixed random seed (42) was also applied during the fivefold cross-validation process, ensuring that fold generation and metric estimation remained reproducible across repeated runs. The performance of the models was compared using key metrics such as accuracy, precision, recall, specificity, and F1-score, averaged over stratified5-fold cross-validation, with mean ± standard deviation reported for each metric. This approach helps provide a more reliable estimate of model performance and determine which model most accurately predicts students at risk of mental health issues. Class imbalance was addressed using stratified k-fold cross-validation and class weighting in Random Forest and SVM. SMOTE was not used due to risk of synthetic bias in psychological data.

3 Results

Table 1 presents the association between depression and various sociodemographic and academic factors. The overall prevalence of depression among the surveyed students was 85.48%, indicating a high burden of depressive symptoms in this population (see Supplementary Table 2 for item-level response distributions). Depression was slightly more prevalent among male students (87.8%) compared to females (81.9%), though the difference was not statistically significant (p = 0.095). While depression rates increased

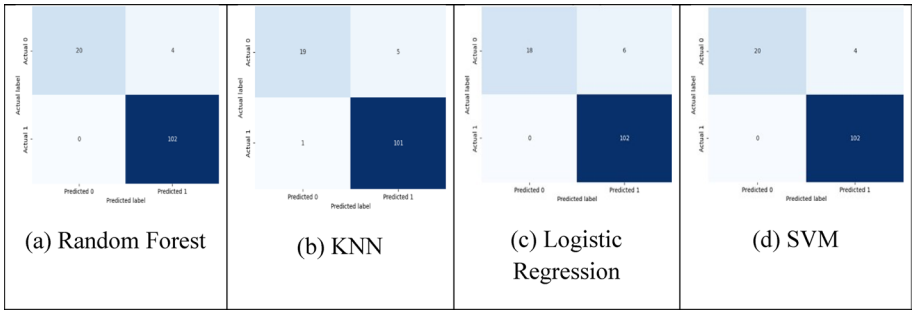


Fig. 1 Confusion Matrices for Each Model Used

Table 1 Association between Depression with Gender, Academic Year, CGPA, and Family Status

Variable	Categories	Depression		P-value*
		No (%)	Yes (%)	
Gender	Female	30 (18.1)	136 (81.9)	0.095
	Male	31 (12.2)	223 (87.8)	
Academic Year	1st	20 (19.6)	82 (80.4)	0.302
	2nd	13 (15.5)	71 (84.5)	
	3rd	13 (13.3)	85 (86.7)	
	4th	15 (11.0)	121 (89.0)	
CGPA	2.5–3.0	1 (1.3)	75 (98.7)	< 0.001*
	3.0–3.5	10 (4.5)	213 (95.5)	
	3.5–3.8	27 (33.8)	53 (66.3)	
	3.8–4.0	23 (56.1)	18 (43.9)	
Family Support	Not Supportive	2 (2.1)	93 (97.9)	< 0.001*
	Moderate	19 (11.4)	148 (88.6)	
	Supportive	40 (25.3)	118 (74.7)	
Relationship Status	Married	2 (7.7)	24 (92.3)	0.307
	Single	59 (15.0)	335 (85.0)	
Financial Expense Bearer	Self-Earner	10 (4.9)	193 (95.1)	< 0.001*
	Supported by Family	51 (23.5)	166 (76.5)	
Experiences Ragging	No	55 (24.2)	172 (75.8)	< 0.001*
	Yes	6 (3.1)	187 (96.9)	
Session Jam	Frequently	5 (1.9)	255 (98.1)	< 0.001*
	Occasionally	56 (35.0)	104 (65.0)	
Social Circle	Large	34 (63.0)	20 (37.0)	< 0.001*
	Moderate	17 (19.3)	71 (80.7)	
	No	39 (66.1)	20 (33.9)	
Adjustment Challenges	Moderate	16 (20.0)	64 (80.0)	< 0.001*
	Yes	6 (2.1)	275 (97.9)	
Mental Health Consultancy	Absent	19 (5.1)	356 (94.9)	< 0.001*
Service in Campus	Present	42 (93.3)	3 (6.7)	< 0.001*
Peer Pressure	No	59 (36.0)	105 (64.0)	< 0.001*
	Yes	2 (0.8)	254 (99.2)	
Educational System Stress	Adjustable	46 (88.5)	6 (11.5)	< 0.001*
	Stressful	15 (4.1)	353 (95.9)	
Total		61(14.5%)	359(85.5%)	

**P*-value calculated from Chi-Square test. Some cells had expected counts < 5; findings from those subgroups were interpreted cautiously

progressively from 1st year (80.4%) to 4th year students (89.0%), no significant association was observed between academic year and depression ($p = 0.302$).

A strong and statistically significant association was observed between depression and CGPA ($p < 0.001$). Students with lower academic performance (CGPA 2.5–3.0) had the highest depression rate (98.7%), while those with higher CGPAs (3.8–4.0) had the lowest (43.9%). Similarly, family support showed a significant association ($p < 0.001$); students who reported unsupportive families experienced much higher depression (97.9%) compared to those who felt supported (74.7%).

Relationship status was not significantly associated with depression ($p = 0.307$). However, students who were financially independent (self-earners) had significantly higher depression (95.1%) compared to those supported by family (76.5%) ($p < 0.001$).

Exposure to ragging and frequent session jams were both significantly linked to higher depression rates ($p < 0.001$). Students who experienced ragging had a 96.9% depression

Table 2 Mean and Standard Deviation of Precision, Recall, Accuracy, F1-score and Specificity of the Classifiers using fivefold cross-validation

Model	Precision	Recall	Accuracy	F1-score	Specificity
Random Forest	0.981 ± 0.018	0.983 ± 0.010	0.969 ± 0.018	0.982 ± 0.010	0.887 ± 0.109
KNN	0.962 ± 0.015	0.983 ± 0.010	0.952 ± 0.013	0.972 ± 0.007	0.772 ± 0.092
Logistic Regression	0.973 ± 0.016	0.989 ± 0.010	0.967 ± 0.009	0.981 ± 0.005	0.837 ± 0.103
SVM	0.984 ± 0.016	0.989 ± 0.010	0.976 ± 0.017	0.986 ± 0.010	0.903 ± 0.096

Table 3 AUC for the test of the Models

Model	AUC	Cross Validation	
		Mean	Standard deviation
KNN	0.917660	0.941322	0.036188
Logistic Regression	0.993499	0.993755	0.005874
Random Forest	0.993499	0.99179	0.009348
SVM	0.995666	0.995055	0.003525

rate versus 75.8% among those who did not. Similarly, students experiencing frequent session jams reported a 98.1% depression rate.

Social connectedness played a notable role. Students with a large social circle reported significantly lower depression (37%) than those with moderate or higher (80.7%) or no social circles (33.9%) ($p < 0.001$). Adjustment challenges also had a significant association ($p < 0.001$), with severe difficulty correlating with a 97.9% depression rate.

Notably, absence of mental health consultancy was strongly linked to depression (94.9%), while its presence showed a significantly lower prevalence (6.7%) ($p < 0.001$). Peer pressure (99.2% vs. 64%) and stress from the educational system (95.9% vs. 11.5%) were also significantly associated with depression (both $p < 0.001$).

3.1 Comparative evaluation of models' performance

Table 2 presents the comparative evaluation of machine learning models used for depression prediction, including Random Forest, K-Nearest Neighbor (KNN), Logistic Regression, and Support Vector Machine (SVM). Among these classifiers, SVM demonstrated the highest accuracy (0.976 with standard deviation 0.017) and F1-score (0.986 with standard deviation 0.01), followed by Random Forest which achieved an accuracy of 0.969 with standard deviation 0.018 and an F1-score of 0.982 with standard deviation 0.01. KNN and Logistic Regression also exhibited strong accuracy (0.952 with standard deviation 0.013 and 0.967 with standard deviation 0.009 respectively). The specificity values of SVM was highest which indicated that this method is better at correctly identifying negative cases.

Among the models evaluated, the SVM achieved the highest balanced performance across precision, recall, specificity, F1-score and AUC under stratified cross-validation, indicating superior ability to identify both depressed and non-depressed students.

To ensure the robustness and generalizability of the model, we performed a thorough assessment of its performance, including an analysis of overfitting and the calculation of key metrics such as AUC of the test data for the Models in Table 3. In addition, ROC curve and learning curves are also demonstrated the high predictive accuracy and strong generalization capability without evidence of overfitting of SVM model (Fig. 2).

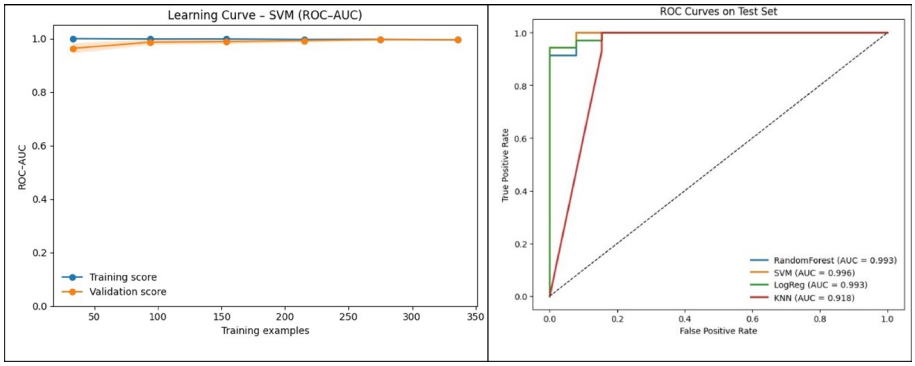
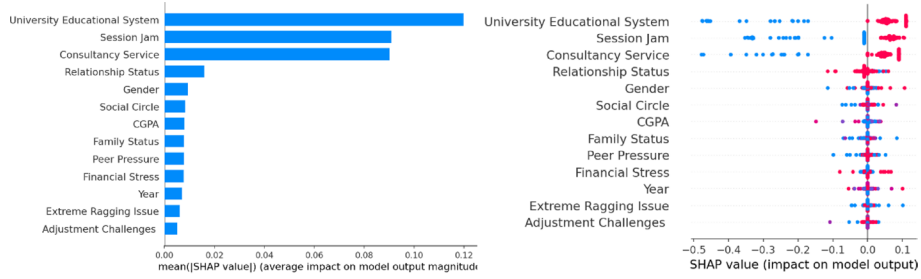


Fig. 2 Learning Curve and ROC Curve from Random forest



(a) Feature Importance **(b) Summary Plot**

Fig. 3 Feature Importance and Impact of Predictors on Depression Prediction

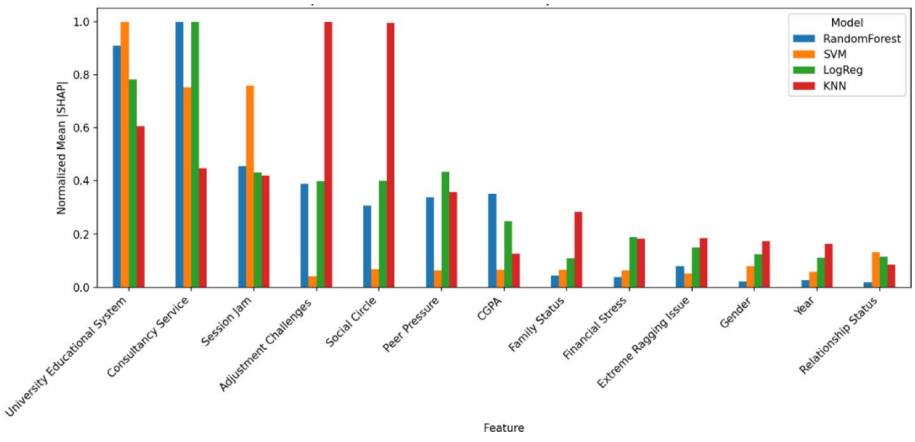


Fig. 4 Mean absolute SHAP values comparing global feature importance across SVM, Logistic Regression, and KNN models, where higher values indicate stronger influence on depression prediction

3.2 Model explainability using SHAP analysis

To further interpret the predictions of our models, we performed SHapley Additive exPlanations (SHAP) analysis on Support Vector Machine (SVM) as it is the best model (Fig. 3). Figure 4 presents a comparison of mean absolute SHAP values across models, providing a global perspective on the most influential features. For more granular interpretability, summary plots (not shown here) were also examined, which confirmed that higher values of 'Consultancy Service' and 'Educational System' consistently reduced predicted risk of depression. Dependence plots were generated for the two

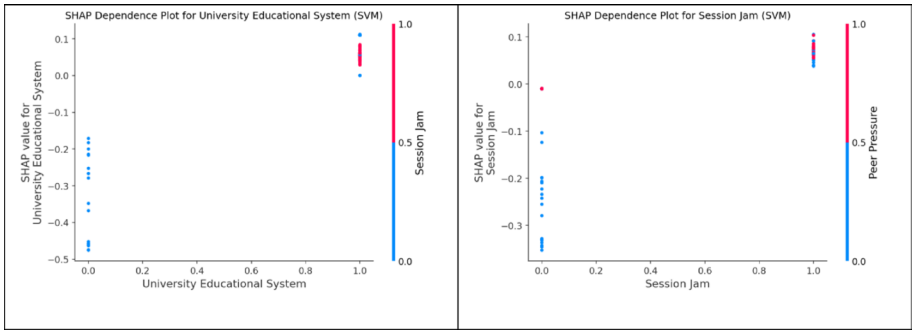


Fig. 5 SHAP Dependence Plot for Top Two Features of SVM

Table 4 Final Hyperparameters Selected for Each Model after Grid Search

Model	Final Hyperparameters Used in Reported Results
Random Forest	n_estimators = 100; criterion = gini; max_depth = 10; min_samples_split = 5
SVM	kernel = linear; C = 1.0; gamma = scale
KNN	n_neighbors = 5; weights = uniform; metric = minkowski; p = 2
Logistic Regression	penalty = l2; C = 1.0; solver = lbfgs; max_iter = 100

most influential features of SVM in Figs. 5. These plots illustrate non-linear relationships between features and predicted outcomes, as well as interactions with secondary features such as Peer Pressure and Adjustment Challenges. To provide a holistic view, Fig. 5 presents a combined comparison of normalized SHAP values across all models. This comparative analysis reveals consistent importance of University Educational System and Consultancy Service across different classifiers, underscoring their robustness as key predictors (Table 4).

3.3 Development of an interface for predicting depression

A user-friendly system interface using SVM was developed to predict students' risk of depression. The system allows users to input various personal attributes such as gender, age, CGPA, family status, financial stress, and social activity engagement. Based on these inputs, the system generates a probability score for depression risk, providing an automated and objective assessment.

3.4 Predictive strategies

Figure 6 highlights solutions addressing financial stress, peer pressure, lack of mental health services, and session jams. Scholarships and financial aid were the most preferred solutions (71.2%), while 95.0% of students emphasized the importance of self-esteem-building activities to counter peer pressure. Additionally, 99.1% of students supported hiring a psychiatrist on campus to improve mental health services.

Figure 7 presents strategies for tackling challenges such as unsupportive families, educational stress, and low CGPA. Open family communication was favored by 73.4% of students as a means to strengthen family support, while 83.8% identified a flexible curriculum as essential for reducing educational stress. To address low CGPA, 92.8% recommended work experience and research involvement as effective measures.

Figure 8 outlines stress and depression management strategies among different student groups. The majority of married students (81.1%) preferred seeking professional help for mental health concerns. Similarly, 81.5% of students emphasized participation

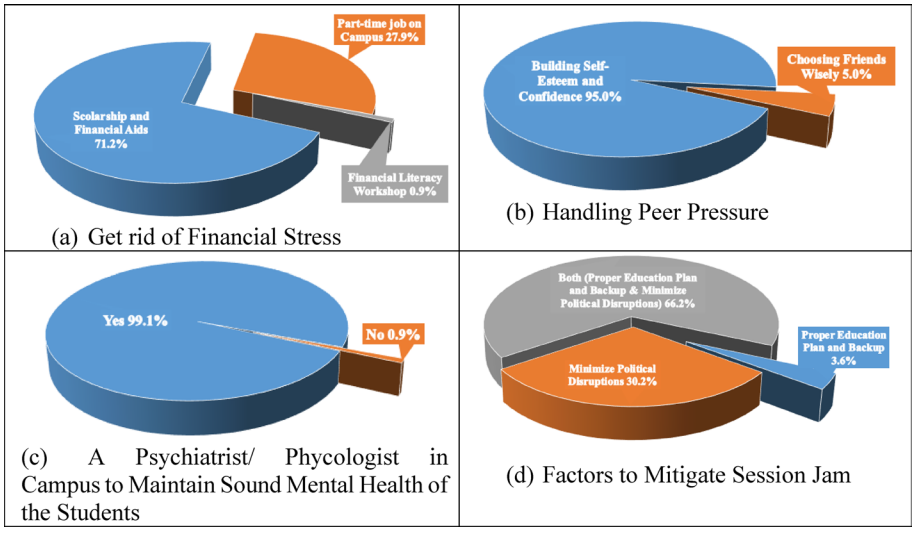


Fig. 6 Strategies for reducing financial stress, managing peer pressure, implementing mental health service on campus, and mitigating session jams

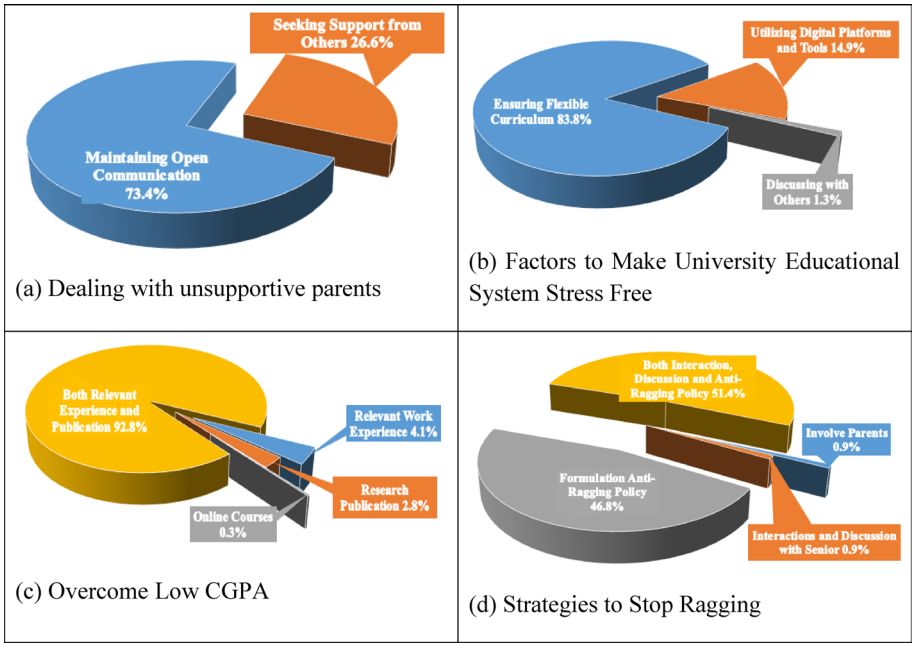


Fig. 7 Strategies for dealing with unsupportive family, making university educational system stress-free, overcoming low CGPA and to stop ragging

in campus clubs as a stress-relief measure, while 95.5% of male students favored exercise and physical activity as their primary stress management approach.

4 Discussion

Findings of this study align with global research on mental health challenges among university students, reaffirming the significant burden of depression and stress in academic environments. Consistent with prior studies in the United States and other high-income countries, the prevalence of depression among students in this study was notably high, with financial stress, academic pressure, and lack of family support emerging as primary

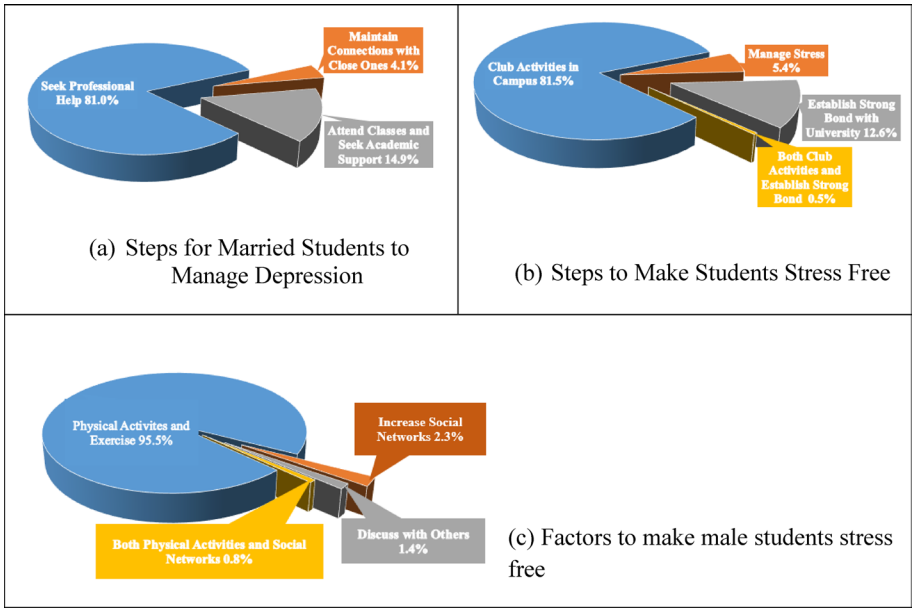


Fig. 8 Suggested strategies for stress and depression management

contributing factors [1, 2]. Similar trends have been observed in other low- and middle-income countries (LMICs), including Bangladesh, where university students frequently report high levels of psychological distress due to socioeconomic and institutional factors [3, 5]. This study further emphasizes the interplay between academic stressors and financial constraints, underscoring the urgent need for targeted interventions to address mental health issues within public universities in Bangladesh.

Our findings mirror Jianwu et al. [15], who reported academic performance as a primary driver of student depression in a large-scale study in China. Rois et al. [9] and Munir et al. [10] similarly found financial stress and social support among the strongest machine-learning–identified predictors in Bangladeshi cohorts. These consistencies underscore the generalizability of our model outputs across LMIC settings. Academic stress has been extensively documented as a key predictor of mental health problems among university students globally [15]. The present study found a strong correlation between low CGPA and depression, reinforcing the idea that academic performance plays a crucial role in shaping students' mental well-being. This finding aligns with previous research conducted in Bangladesh, where students with lower academic achievements exhibited higher depression rates [6]. Additionally, session jams, a common issue in Bangladeshi public universities, were significantly associated with higher depression levels. Prolonged academic delays and uncertainties regarding graduation create significant stress among students, similar to findings in other LMICs where inadequate educational infrastructure exacerbates mental health risks [16]. Addressing academic stress through curriculum modifications, flexible grading systems, and improved academic support structures could play a critical role in mitigating mental health challenges.

Social support networks have consistently been highlighted as protective factors against depression among students [7]. This study revealed that students with strong social connections had significantly lower depression rates, which is consistent with previous research emphasizing the positive impact of peer support on mental health outcomes [8]. In contrast, students experiencing social isolation exhibited significantly

higher depression levels. This finding underscores the need for universities to foster peer engagement initiatives, mentorship programs, and student-led mental health support groups to enhance psychological resilience among students.

Financial constraints were another critical determinant of depression in this study, with students experiencing financial stress reporting significantly higher mental health issues. These findings align with studies in both high-income and LMIC contexts, where financial insecurity has been identified as a major contributor to student mental health problems [3]. Public university students in Bangladesh, often from lower-income backgrounds, face substantial economic challenges, including tuition costs, accommodation expenses, and limited access to financial aid. In comparison, studies conducted in Western countries highlight similar financial burdens, albeit with more structured financial aid programs and student counseling services to mitigate these stressors [1]. Implementing targeted financial assistance programs, scholarships, and part-time job opportunities within university systems could help alleviate financial stress and improve students' overall well-being.

Another notable finding was the significant association between exposure to ragging and depression. The negative impact of ragging on students' mental health has been well-documented in South Asian contexts, particularly in India and Bangladesh, where hazing practices persist despite institutional policies [5]. The high prevalence of depression among students who had experienced ragging indicates the urgent need for stricter enforcement of anti-ragging policies, awareness campaigns, and student counseling services to prevent psychological distress stemming from such experiences.

Access to mental health services was a significant protective factor in this study, with students who had access to counseling services reporting lower levels of depression. This finding is consistent with global literature emphasizing the role of mental health services in mitigating student distress [16]. However, mental health services remain largely inaccessible or underutilized in Bangladeshi public universities due to stigma, lack of awareness, and limited institutional support. Cultural stigma around mental health in Bangladesh also may influence both underreporting and heightened distress due to societal expectations. Studies from developed countries demonstrate the effectiveness of integrated mental health programs within university settings, including digital mental health interventions, peer counseling, and university-based psychotherapy services [8]. Implementing similar initiatives in Bangladeshi public universities could substantially improve students' mental health outcomes.

The study's findings also reinforce the potential of machine learning in predicting mental health risks among students. Among the models tested, Random Forest and Support Vector Machine (SVM) exhibited the highest predictive accuracy, which aligns with previous studies demonstrating the efficacy of these models in identifying individuals at risk of depression. National estimates from WHO suggest depression prevalence in general Bangladeshi adult populations is 4.6%, highlighting the elevated risk in university cohorts [9, 10]. The development of a predictive interface for identifying at-risk students represents a significant advancement in mental health research, offering a scalable and data-driven approach to early intervention efforts. Feature importance was assessed using SHAP values, indicating CGPA, family support, financial stress, and session jams as top predictors.

4.1 Limitations and implications

Despite its contributions, this study has certain limitations. Its cross-sectional design prevents causal inference, necessitating longitudinal research for deeper insights. The sample size of 420, while sufficient for preliminary model training and cross-validation, may limit generalizability. Larger, multicentric datasets are recommended for future research.The high observed prevalence of depressive symptoms may reflect this self-selection bias and must be interpreted accordingly. Future studies should adopt stratified, in-person, or mixed-mode recruitment strategies to improve representativeness. The focus on public university students limits generalizability; research on private university students is needed for comparison. Additionally, while machine learning models showed high accuracy, concerns regarding data privacy, algorithmic bias, and real-world application require further examination. Use of self-reported questionnaires and lack of diagnostic clinical evaluation may limit accuracy of depression categorization. Potential algorithmic bias due to demographic skews or data imbalance was considered; fairness metrics can be explored in future work. Another limitation relates to the statistical comparisons of model performance. We applied independent t-tests to evaluate differences among classifiers, recognizing that cross-validation folds are not entirely independent. While our findings were consistent across folds and metrics, future studies with larger datasets should adopt more rigorous comparative methods to provide stronger statistical confidence. False positives may lead to unnecessary interventions, while false negatives may leave students undiagnosed. These trade-offs require careful threshold selection. Addressing these limitations can enhance understanding and intervention strategies for student mental health.

Based on these findings, universities should prioritize expanding mental health resources, integrating mental health education into curricula, and fostering supportive peer networks to mitigate student distress. Establishing financial aid programs and enforcing strict anti-ragging policies could further enhance student well-being. Future research should explore the long-term effectiveness of predictive models in real-world university settings to refine and optimize mental health interventions, while also incorporating standardized instruments like PHQ-9 and DASS-21, rather than bespoke questionnaires to enhance validity and comparability. Additionally, future work could leverage SHAP or other explainability frameworks to further dissect model behavior and guide the design of personalized interventions.

5 Conclusion

This study highlights the impact of socio-economic and academic factors on student mental health, particularly depression, supporting the study's objectives and hypothesis. Findings show higher depression rates among male and final-year students; while family support is protective and financial stress exacerbates distress, the availability of adequately resourced student-consultancy/counselling services and a well-designed university education system is crucial for prevention and timely support. The developed depression prediction system offers early identification and intervention opportunities. To address student mental health challenges, universities should implement awareness programs, accessible counseling, and financial aid support. However, limitations such as a small sample size and self-reported data suggest the need for future research incorporating diverse populations, qualitative insights, and real-time predictive

models to enhance understanding and intervention strategies for student mental health in Bangladesh.

Supplementary Information

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Supplementary Material 1

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Author contributions
S.B. conceptualized the study, conducted data curation, formal analysis, investigation, methodology, visualization, and wrote the original draft. M.F.H. contributed to methodology, supervision, and manuscript review and editing. S.A.A. was responsible for visualization and manuscript review and editing. M.M.H.S. contributed to methodology, validation, visualization, original draft writing, and manuscript review and editing. M.K.H. supervised the study, contributed to methodology, and participated in manuscript review and editing. All authors reviewed and approved the final manuscript.

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Availability of data and materials
The datasets and code of analysis are available in the GitHubrepository, https://github.com/sudhabh23/Mental_Health_Prevention_Bangladeshi_Students.

Declarations

Ethics approval and consent to participate
The study received ethical approval from the Research Ethics Committee of the Faculty of Health and Life Sciences, Daffodil International University, Dhaka, Bangladesh (Approval Reference: FHLSREC/DIU/2024/SMIG-141). Informed consent was obtained from all participants before data collection. Respondents were informed about the study's objectives, data confidentiality, and their right to withdraw at any stage without consequences. Data were anonymized to protect the privacy of participants. The study adhered to the ethical principles of the Declaration of Helsinki and ensured no coercion or exploitation of vulnerable groups. No experiments on humans and/or the use of human tissue samples were done in this study.

Competing interests
The authors declare no competing interests.

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